Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

Integrals

1.1 Medium Integral

Trigonometric Rational Integral

Evaluate the definite integral:

$$\int_0^{\pi/6} \frac{\tan(2x)}{\sqrt{\sin^6(x)+\cos^6(x)}}\,dx = \ln\left(\frac{a+\sqrt{b}}{c}\right)$$

where a, b, c are co-prime natural numbers. Find $10ac + b$.

Solution: We begin by simplifying the algebraic structure of the denominator. Using the sum of cubes factorization $u^3 + v^3 = (u+v)(u^2 - uv + v^2)$ with $u = \sin^2(x)$ and $v = \cos^2(x)$:

$$\sin^6(x)+\cos^6(x) = \left(\sin^2(x)+\cos^2(x)\right)\left(\sin^4(x)-\sin^2(x)\cos^2(x)+\cos^4(x)\right)$$

Since the fundamental Pythagorean identity gives $\sin^2(x)+\cos^2(x) = 1$, we rewrite the remaining quadratic terms:

$$\begin{aligned}\sin^4(x)-\sin^2(x)\cos^2(x)+\cos^4(x) &= \left(\sin^2(x)+\cos^2(x)\right)^2 - 3\sin^2(x)\cos^2(x)\\ &= 1-3\sin^2(x)\cos^2(x)\end{aligned}$$

Applying the double-angle identity $\sin(2x) = 2\sin(x)\cos(x) \implies \sin^2(x)\cos^2(x) = \frac{1}{4}\sin^2(2x)$, we obtain:

$$\sin^6(x)+\cos^6(x) = 1-\frac{3}{4}\sin^2(2x)$$

We also express $\sin^2(2x)$ in terms of $\cos(2x)$ via $\sin^2(2x) = 1-\cos^2(2x)$:

$$1-\frac{3}{4}\left(1-\cos^2(2x)\right) = \frac{1}{4}+\frac{3}{4}\cos^2(2x) = \frac{1+3\cos^2(2x)}{4}$$

Taking the square root of this simplified expression transforms the denominator into:

$$\sqrt{\sin^6(x)+\cos^6(x)} = \frac{\sqrt{1+3\cos^2(2x)}}{2}$$

Substituting this back into the original definite integral and writing $\tan(2x) = \frac{\sin(2x)}{\cos(2x)}$:

$$I = \int_0^{\pi/6} \frac{\frac{\sin(2x)}{\cos(2x)}}{\frac{\sqrt{1+3\cos^2(2x)}}{2}}\,dx = \int_0^{\pi/6} \frac{2\sin(2x)}{\cos(2x)\sqrt{1+3\cos^2(2x)}}\,dx$$

We now introduce the substitution $u = \cos(2x)$, which gives $du = -2\sin(2x)\,dx \implies 2\sin(2x)\,dx = -du$. We transform the limits of integration accordingly:

- **Lower limit ($x = 0$):** $u = \cos(0) = 1$.
- **Upper limit ($x = \pi/6$):** $u = \cos(\pi/3) = \frac{1}{2}$.

Substituting u, du, and the new limits reverses the orientation of integration:

$$I = \int_1^{1/2} \frac{-1}{u\sqrt{1+3u^2}}\,du = \int_{1/2}^1 \frac{1}{u\sqrt{1+3u^2}}\,du$$

To evaluate this integral, we make a second algebraic substitution by setting $v = \sqrt{1+3u^2}$. Squaring both sides yields $v^2 = 1+3u^2 \implies u^2 = \frac{v^2-1}{3}$. Differentiating implicitly with respect to u gives $2v\,dv = 6u\,du \implies \frac{du}{u} = \frac{v\,dv}{3u^2} = \frac{v\,dv}{v^2-1}$. We update the integration boundaries for the variable v:

- When $u = \frac{1}{2}$: $v = \sqrt{1+3\left(\frac{1}{4}\right)} = \sqrt{\frac{7}{4}} = \frac{\sqrt{7}}{2}$.
- When $u = 1$: $v = \sqrt{1+3(1)} = \sqrt{4} = 2$.

The integral simplifies to a standard rational form:

$$I = \int_{\sqrt{7}/2}^{2} \frac{1}{v^2-1}\,dv$$

Using partial fraction decomposition $\frac{1}{v^2-1} = \frac{1}{2}\left(\frac{1}{v-1} - \frac{1}{v+1}\right)$, we integrate term by term:

$$I = \frac{1}{2}\left[\ln\left|\frac{v-1}{v+1}\right|\right]_{\sqrt{7}/2}^{2}$$

Evaluating at the upper limit $v = 2$:

$$\frac{1}{2}\ln\left(\frac{2-1}{2+1}\right) = \frac{1}{2}\ln\left(\frac{1}{3}\right) = -\frac{1}{2}\ln(3) = \ln\left(\frac{1}{\sqrt{3}}\right)$$

Evaluating at the lower limit $v = \frac{\sqrt{7}}{2}$:

$$\frac{1}{2}\ln\left(\frac{\frac{\sqrt{7}}{2}-1}{\frac{\sqrt{7}}{2}+1}\right) = \frac{1}{2}\ln\left(\frac{\sqrt{7}-2}{\sqrt{7}+2}\right)$$

We rationalize the denominator inside the logarithm by multiplying by the conjugate $(\sqrt{7}-2)$:

$$\frac{\sqrt{7}-2}{\sqrt{7}+2}\cdot\frac{\sqrt{7}-2}{\sqrt{7}-2} = \frac{(\sqrt{7}-2)^2}{7-4} = \frac{(\sqrt{7}-2)^2}{3}$$

$$\frac{1}{2}\ln\left(\frac{(\sqrt{7}-2)^2}{3}\right) = \ln\left(\frac{\sqrt{7}-2}{\sqrt{3}}\right)$$

Subtracting the lower evaluation from the upper evaluation:

$$I = \ln\left(\frac{1}{\sqrt{3}}\right) - \ln\left(\frac{\sqrt{7}-2}{\sqrt{3}}\right) = \ln\left(\frac{1}{\sqrt{7}-2}\right)$$

Rationalizing the final expression:

$$\frac{1}{\sqrt{7}-2} \cdot \frac{\sqrt{7}+2}{\sqrt{7}+2} = \frac{2+\sqrt{7}}{3}$$

Therefore, the exact value of the integral is:

$$I = \ln\left(\frac{2+\sqrt{7}}{3}\right)$$

Comparing this result with the form $\ln\left(\frac{a+\sqrt{b}}{c}\right)$, we identify:

$$a = 2, \qquad b = 7, \qquad c = 3$$

We verify that $\gcd(2,7,3) = 1$, confirming they are co-prime natural numbers. Finally, we compute the requested combination:

$$10ac + b = 10(2)(3) + 7 = 60 + 7 = 67$$

The exact value is:

$$67$$

1.2 Hard Integral

Floor Function Inverse Square Root Integral

Evaluate the definite integral:

$$\int_0^{1/9} \left\lfloor \sqrt{\frac{1}{x} - 9} \right\rfloor \, dx$$

Solution: Let the target integral over the interval $\left(0, \frac{1}{9}\right]$ be denoted by I. Let us analyze the behavior of the algebraic function inside the floor operator:

$$y(x) = \sqrt{\frac{1}{x} - 9}$$

When $x = \frac{1}{9}$, the function vanishes: $y\left(\frac{1}{9}\right) = \sqrt{9-9} = 0$. As $x \to 0^+$, the reciprocal $\frac{1}{x} \to \infty$, causing $y(x)$ to grow monotonically without bound toward $+\infty$. Since $y(x)$ is strictly decreasing on the interval $\left(0, \frac{1}{9}\right]$, the floor function $\lfloor y(x) \rfloor$ takes constant non-negative integer values $n \in \mathbb{Z}_{\geq 0}$ across consecutive subintervals.

We determine the precise subinterval boundaries where $\lfloor y(x) \rfloor = n$ by solving the inequality:

$$n \leq \sqrt{\frac{1}{x} - 9} < n + 1$$

Squaring all terms (since $n \geq 0$) and adding 9:

$$n^2 + 9 \leq \frac{1}{x} < (n+1)^2 + 9$$

Taking the reciprocal reverses the inequalities:

$$\frac{1}{(n+1)^2 + 9} < x \leq \frac{1}{n^2 + 9}$$

Let us define the sequence of boundary points x_n for $n \in \mathbb{Z}_{\geq 0}$ by:

$$x_n = \frac{1}{n^2 + 9}$$

Notice that $x_0 = \frac{1}{0^2+9} = \frac{1}{9}$ corresponds precisely to the upper limit of our integral. On each subinterval $x \in (x_{n+1}, x_n]$, the integrand evaluates to the constant integer n. We can therefore partition the integral into an infinite sum of rectangular areas:

$$I = \sum_{n=0}^{\infty} \int_{x_{n+1}}^{x_n} n \, dx = \sum_{n=0}^{\infty} n \left(x_n - x_{n+1} \right)$$

Since the term corresponding to $n = 0$ evaluates to zero ($0 \cdot (x_0 - x_1) = 0$), the summation begins effectively at $n = 1$:

$$I = \sum_{n=1}^{\infty} n \left(x_n - x_{n+1} \right)$$

We evaluate this infinite series by considering the N-th partial sum and applying summation by parts (telescoping rearrangement):

$$S_N = \sum_{n=1}^{N} n \left(x_n - x_{n+1} \right) = 1(x_1 - x_2) + 2(x_2 - x_3) + 3(x_3 - x_4) + \cdots + N(x_N - x_{N+1})$$

Regrouping terms by the sequence elements x_n:

$$S_N = x_1 + x_2 + x_3 + \cdots + x_N - N x_{N+1} = \sum_{n=1}^{N} x_n - N x_{N+1}$$

We analyze the asymptotic behavior of the boundary term $N x_{N+1}$ as $N \to \infty$:

$$\lim_{N \to \infty} N x_{N+1} = \lim_{N \to \infty} \frac{N}{(N+1)^2 + 9} = \lim_{N \to \infty} \frac{N}{N^2 + 2N + 10} = 0$$

Since the boundary term vanishes in the limit, the integral reduces directly to the sum of the boundary points:

$$I = \sum_{n=1}^{\infty} x_n = \sum_{n=1}^{\infty} \frac{1}{n^2 + 9}$$

To compute the exact closed-form sum of this series, we utilize the classic Mittag-Leffler partial fraction expansion for the hyperbolic cotangent function:

$$\pi x \coth(\pi x) = 1 + 2x^2 \sum_{n=1}^{\infty} \frac{1}{n^2 + x^2}$$

Isolating the series sum on the right-hand side yields the general identity:

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + x^2} = \frac{\pi x \coth(\pi x) - 1}{2x^2}$$

In our specific series, we substitute $x^2 = 9 \implies x = 3$:

$$I = \sum_{n=1}^{\infty} \frac{1}{n^2 + 3^2} = \frac{3\pi \coth(3\pi) - 1}{2(3^2)} = \frac{3\pi \coth(3\pi) - 1}{18}$$

The exact analytical value of the definite integral is:

$$\int_0^{1/9} \left\lfloor \sqrt{\frac{1}{x} - 9} \right\rfloor dx = \frac{3\pi \coth(3\pi) - 1}{18}$$

Differentials

2.1 Medium Differential

Implicit Logarithmic-Arc Tangent Derivative

Given the function defined implicitly by:

$$\tan^{-1}\left(\frac{y}{x}\right) = \ln\left(\sqrt{x^2+y^2}\right)$$

Find the exact value of the derivative $\frac{dy}{dx}$ at the point $(\ln(6), \ln(2))$.

Solution: We begin by simplifying the right-hand side of the implicit equation using standard logarithmic properties:

$$\tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\ln\left(x^2+y^2\right)$$

We differentiate both sides of this relation implicitly with respect to the independent variable x. For the left-hand side, we apply the chain rule along with the quotient rule for the argument $\frac{y}{x}$:

$$\frac{d}{dx}\left[\tan^{-1}\left(\frac{y}{x}\right)\right] = \frac{1}{1+\left(\frac{y}{x}\right)^2}\cdot\frac{d}{dx}\left(\frac{y}{x}\right) = \frac{x^2}{x^2+y^2}\cdot\frac{x\frac{dy}{dx}-y}{x^2} = \frac{x\frac{dy}{dx}-y}{x^2+y^2}$$

For the right-hand side, we apply the chain rule to the natural logarithm:

$$\frac{d}{dx}\left[\frac{1}{2}\ln\left(x^2+y^2\right)\right] = \frac{1}{2}\cdot\frac{1}{x^2+y^2}\cdot\left(2x+2y\frac{dy}{dx}\right) = \frac{x+y\frac{dy}{dx}}{x^2+y^2}$$

Equating the derivatives of both sides:

$$\frac{x\frac{dy}{dx}-y}{x^2+y^2} = \frac{x+y\frac{dy}{dx}}{x^2+y^2}$$

Since $x^2+y^2 \neq 0$, we multiply through by the common denominator (x^2+y^2):

$$x\frac{dy}{dx} - y = x + y\frac{dy}{dx}$$

We now group all terms involving the derivative $\frac{dy}{dx}$ on the left-hand side and move the remaining terms to the right-hand side:

$$x\frac{dy}{dx} - y\frac{dy}{dx} = x+y \implies (x-y)\frac{dy}{dx} = x+y$$

Solving explicitly for the derivative:

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

We evaluate this general derivative expression at the given point $(x_0, y_0) = (\ln(6), \ln(2))$:

$$\left.\frac{dy}{dx}\right|_{(\ln(6),\ln(2))} = \frac{\ln(6) + \ln(2)}{\ln(6) - \ln(2)}$$

Using the fundamental product and quotient rules of logarithms:

$$\ln(6) + \ln(2) = \ln(6 \times 2) = \ln(12)$$

$$\ln(6) - \ln(2) = \ln\left(\frac{6}{2}\right) = \ln(3)$$

Substituting these simplified logarithmic values:

$$\frac{dy}{dx} = \frac{\ln(12)}{\ln(3)}$$

By the change-of-base formula for logarithms, this can be written compactly as:

$$\frac{dy}{dx} = \log_3(12) = \log_3(3 \times 4) = 1 + 2\log_3(2)$$

The exact value of the derivative is:

$$\frac{\ln(12)}{\ln(3)}$$

2.2 Hard Differential

Third-Order Partial Differential Equation

A function $u(x, y)$ satisfies the following PDE in the first quadrant:

$$\frac{\partial^3 u}{\partial x^2 \partial y} - y^2 \frac{\partial u}{\partial y} + y^3 = 0$$

Knowing that $u(x, 0) = -\frac{x^2}{2}$, $\frac{\partial u}{\partial y}(1, y) = e^{-y} + y$ and $\lim_{x\to\infty} \frac{\partial^2 u}{\partial x \partial y} = 0$ for all $y > 0$, find $u(1, 1)$.

Solution: We observe that the dependent variable $u(x, y)$ appears in the partial differential equation exclusively within derivatives with respect to y. Let us define an auxiliary function $v(x, y)$ representing the first partial derivative of u with respect to y:

$$v(x, y) = \frac{\partial u}{\partial y}(x, y)$$

In terms of $v(x, y)$, the third-order PDE transforms into a second-order equation:

$$\frac{\partial^2 v}{\partial x^2} - y^2 v + y^3 = 0 \implies \frac{\partial^2 v}{\partial x^2} - y^2 v = -y^3$$

Since all differentiations in this transformed equation are with respect to x, we can treat $y > 0$ as a constant parameter and solve it as a second-order linear ordinary differential equation with constant coefficients for $v(x, y)$ as a function of x.

The associated homogeneous differential equation is:

$$\frac{\partial^2 v}{\partial x^2} - y^2 v = 0$$

The characteristic equation is $r^2 - y^2 = 0$, yielding roots $r = \pm y$. Therefore, the complementary solution is:

$$v_c(x, y) = A(y)e^{-yx} + B(y)e^{yx}$$

where $A(y)$ and $B(y)$ are arbitrary functions of the parameter y. For the particular solution $v_p(x, y)$, since the forcing term $-y^3$ is independent of x, we try a constant particular solution $v_p(x, y) = C(y)$:

$$0 - y^2 C(y) = -y^3 \implies C(y) = y$$

Thus, the general solution for $v(x, y)$ is given by:

$$v(x, y) = \frac{\partial u}{\partial y} = A(y)e^{-yx} + B(y)e^{yx} + y$$

We now apply the given boundary and asymptotic conditions to determine the unknown coefficient functions $A(y)$ and $B(y)$. First, consider the asymptotic condition as $x \to \infty$ for $y > 0$:

$$\lim_{x\to\infty} \frac{\partial^2 u}{\partial x \partial y} = \lim_{x\to\infty} \frac{\partial v}{\partial x} = 0$$

Differentiating our expression for $v(x, y)$ with respect to x:

$$\frac{\partial v}{\partial x} = -yA(y)e^{-yx} + yB(y)e^{yx}$$

Since $y > 0$, the exponential term $e^{-yx} \to 0$ as $x \to \infty$. However, the term $e^{yx} \to \infty$ unconditionally unless its coefficient vanishes identically. To prevent divergence and satisfy the zero limit, we must have:

$$B(y) = 0 \quad \text{for all } y > 0$$

This reduces our first derivative expression to:

$$v(x, y) = \frac{\partial u}{\partial y} = A(y)e^{-yx} + y$$

Next, we apply the boundary condition at $x = 1$:

$$\frac{\partial u}{\partial y}(1, y) = e^{-y} + y$$

Substituting $x = 1$ into our reduced expression for $v(x, y)$:

$$v(1, y) = A(y)e^{-y} + y$$

Equating the two expressions for $v(1,y)$:

$$A(y)e^{-y} + y = e^{-y} + y \implies A(y)e^{-y} = e^{-y} \implies A(y) = 1$$

Therefore, the exact partial derivative $\frac{\partial u}{\partial y}$ across the entire first quadrant is:

$$\frac{\partial u}{\partial y}(x,y) = e^{-yx} + y$$

To recover the function $u(x,y)$, we integrate $\frac{\partial u}{\partial y}$ with respect to y:

$$u(x,y) = \int \left(e^{-yx} + y\right) dy = -\frac{1}{x}e^{-yx} + \frac{1}{2}y^2 + f(x)$$

where $f(x)$ is an arbitrary function of x representing the constant of integration with respect to y. We determine $f(x)$ using the initial boundary condition along the x-axis ($y = 0$):

$$u(x,0) = -\frac{x^2}{2}$$

Setting $y = 0$ in our integrated expression for $u(x,y)$:

$$u(x,0) = -\frac{1}{x}e^0 + 0 + f(x) = -\frac{1}{x} + f(x)$$

Equating this to the given initial profile:

$$-\frac{1}{x} + f(x) = -\frac{x^2}{2} \implies f(x) = \frac{1}{x} - \frac{x^2}{2}$$

Substituting $f(x)$ back into our solution yields the complete, exact analytical expression for $u(x,y)$:

$$u(x,y) = -\frac{1}{x}e^{-yx} + \frac{1}{2}y^2 + \frac{1}{x} - \frac{x^2}{2} = \frac{1 - e^{-yx}}{x} + \frac{y^2 - x^2}{2}$$

Finally, we evaluate this function at the point $(x,y) = (1,1)$:

$$u(1,1) = \frac{1 - e^{-(1)(1)}}{1} + \frac{1^2 - 1^2}{2} = 1 - e^{-1} + 0 = 1 - \frac{1}{e}$$

The exact value is:

$$u(1,1) = 1 - \frac{1}{e}$$

Derivatives

3.1 Medium Derivative

Infinite Nested Inverse Sine Composition

Compute $f'\left(\frac{2}{\pi}\right)$ if

$$f(x) = \sin^{-1}\left(x\sin^{-1}\left(x\sin^{-1}(x\dots)\right)\right).$$

Solution: Let the infinite nested composition be denoted by the dependent variable $y = f(x)$:

$$y = \sin^{-1}\left(x\sin^{-1}\left(x\sin^{-1}(x\dots)\right)\right)$$

Taking the sine of both sides of the equation maps the inverse trigonometric relation into an algebraic form:

$$\sin(y) = x\sin^{-1}\left(x\sin^{-1}(x\dots)\right)$$

We observe the self-similar recursive architecture of the expression: the infinite composition appearing inside the parentheses on the right-hand side is precisely our original definition of y. Substituting y back into the right-hand side yields the compact implicit relation governing the function:

$$\sin(y) = xy$$

We first determine the specific evaluation value $y_0 = f\left(\frac{2}{\pi}\right)$ corresponding to $x_0 = \frac{2}{\pi}$. Substituting $x_0 = \frac{2}{\pi}$ into our implicit equation gives:

$$\sin(y_0) = \frac{2}{\pi}y_0 \implies \frac{\sin(y_0)}{y_0} = \frac{2}{\pi}$$

We recognize immediately that $y_0 = \frac{\pi}{2}$ satisfies this relation exactly, since:

$$\frac{\sin(\pi/2)}{\pi/2} = \frac{1}{\pi/2} = \frac{2}{\pi}$$

Since the principal branch of the inverse sine function maps positive arguments to positive angles in the first quadrant, $y_0 = \frac{\pi}{2}$ is the unique valid evaluation point.

We now differentiate the implicit relation $\sin(y) = xy$ with respect to the independent variable x, applying the product rule to the right-hand side:

$$\cos(y)\cdot\frac{dy}{dx} = y + x\frac{dy}{dx}$$

Rearranging terms to isolate the first derivative $\frac{dy}{dx}$:

$$\cos(y)\frac{dy}{dx} - x\frac{dy}{dx} = y \implies (\cos(y) - x)\frac{dy}{dx} = y$$

$$f'(x) = \frac{dy}{dx} = \frac{y}{\cos(y) - x}$$

Substituting our evaluation coordinates $(x_0, y_0) = \left(\frac{2}{\pi}, \frac{\pi}{2}\right)$ directly into this derivative formula:

$$f'\left(\frac{2}{\pi}\right) = \frac{\frac{\pi}{2}}{\cos\left(\frac{\pi}{2}\right) - \frac{2}{\pi}} = \frac{\frac{\pi}{2}}{0 - \frac{2}{\pi}} = \frac{\frac{\pi}{2}}{-\frac{2}{\pi}} = -\frac{\pi^2}{4}$$

The exact value of the derivative is:

$$-\frac{\pi^2}{4}$$

3.2 Hard Derivative

Infinite Nested Radical Inverse Derivative

Compute $(f^{-1}(2))'$, derivative of inverse at $x = 2$, if

$$f(x) = \sqrt{1 + \sqrt{x^2 + \sqrt{x^6 + \sqrt{x^{14} + \cdots}}}}$$

Solution: Let $y = f(x) = \sqrt{1 + \sqrt{x^2 + \sqrt{x^6 + \sqrt{x^{14} + \cdots}}}}$. We begin by squaring both sides of the equation to eliminate the outermost square root:

$$y^2 = 1 + \sqrt{x^2 + \sqrt{x^6 + \sqrt{x^{14} + \cdots}}}$$

Subtracting 1 from both sides isolates the remaining nested radical structure:

$$y^2 - 1 = \sqrt{x^2 + \sqrt{x^6 + \sqrt{x^{14} + \cdots}}}$$

We examine the powers of x inside the consecutive radical layers: the exponents follow the sequence $2, 6, 14, 30, \ldots$, which is generated by the formula $a_n = 2^{n+1} - 2$ for $n \geq 1$. Let us factor out x^2 from under the first square root on the right-hand side:

$$y^2 - 1 = \sqrt{x^2\left(1 + \frac{1}{x^2}\sqrt{x^6 + \sqrt{x^{14} + \cdots}}\right)} = x\sqrt{1 + \frac{1}{x^2}\sqrt{x^6 + \sqrt{x^{14} + \cdots}}}$$

We now bring the factor $\frac{1}{x^2}$ inside the second square root by squaring it to $\frac{1}{x^4}$:

$$y^2 - 1 = x\sqrt{1 + \sqrt{\frac{x^6}{x^4} + \frac{1}{x^4}\sqrt{x^{14} + \cdots}}} = x\sqrt{1 + \sqrt{x^2 + \frac{1}{x^4}\sqrt{x^{14} + \cdots}}}$$

Continuing this process inductively, bringing $\frac{1}{x^4}$ into the third square root as $\frac{1}{x^8}$ reduces x^{14} to x^6. At each successive layer, the division by x^{2^n} shifts the exponent $(2^{n+2} - 2) - 2^{n+1} = 2^{n+1} - 2$, perfectly regenerating the exact original sequence of nested radicals:

$$y^2 - 1 = x\sqrt{1 + \sqrt{x^2 + \sqrt{x^6 + \cdots}}}$$

Since the radical expression multiplying x is precisely our original definition of $y = f(x)$, we obtain the remarkable algebraic identity:

$$y^2 - 1 = xy$$

We can now express the independent variable x explicitly as a function of y:

$$x = \frac{y^2 - 1}{y} = y - \frac{1}{y}$$

This expression is precisely the analytical formula for the inverse function $g(y) = f^{-1}(y)$:

$$f^{-1}(y) = y - y^{-1}$$

To find the derivative of the inverse function evaluated at $y = 2$, we differentiate $g(y)$ with respect to y:

$$\left(f^{-1}(y)\right)' = \frac{d}{dy}\left(y - y^{-1}\right) = 1 + \frac{1}{y^2}$$

Substituting the target evaluation point $y = 2$:

$$\left(f^{-1}(2)\right)' = 1 + \frac{1}{2^2} = 1 + \frac{1}{4} = \frac{5}{4}$$

The exact value of the derivative is:

$$\frac{5}{4}$$

Physics & Engineering

4.1 Mechanics

Composite Solid Center of Mass Calculation

An object on the ground consists of a cylinder A, a cube B and a cone C. The dimensions are given as follows: Cylinder A: diameter $l_A = 4$, height $h_A = \frac{3}{2} l_A$. Cube B: side length $l_B = l_A$, height $h_B = l_B$. Cone C: diameter $l_C = l_B$, height $h_C = \frac{3}{2} h_A$. The cube is on top of the cylinder, and the cone is attached to the side of the cube. Find $n = 2(\bar{x} + \bar{y}) + 1$ to 5 decimal places where $(\bar{x}, \bar{y})$ is the center of mass relative to the cylinder's axis of symmetry and the ground.

Solution: Let us establish a Cartesian coordinate system where the ground lies in the horizontal plane $y = 0$, and the vertical axis of symmetry of Cylinder A corresponds to the line $x = 0$. Assuming uniform mass density across all three geometric components, the center of mass coordinates are determined purely by volumetric weighting. We calculate the geometric properties of each constituent shape:

1. **Cylinder A:**
   - **Dimensions:** Diameter $l_A = 4 \implies$ radius $r_A = 2$. Height $h_A = \frac{3}{2}(4) = 6$.
   - **Position:** Resting on the ground from $y = 0$ to $y = 6$, centered on $x = 0$.
   - **Center of mass** $(\bar{x}_A, \bar{y}_A)$**:** Located on the axis of symmetry at mid-height:
   $$(\bar{x}_A, \bar{y}_A) = \left(0, \frac{6}{2}\right) = (0, 3)$$
   - **Volume V_A:**
   $$V_A = \pi r_A^2 h_A = \pi(2^2)(6) = 24\pi$$
2. **Cube B:**
   - **Dimensions:** Side length $l_B = 4$, height $h_B = 4$.
   - **Position:** Resting symmetrically on top of Cylinder A, spanning vertically from $y = 6$ to $y = 10$, and horizontally from $x = -2$ to $x = +2$.
   - **Center of mass** $(\bar{x}_B, \bar{y}_B)$**:** Centered horizontally at $x = 0$ and vertically at mid-height:
   $$(\bar{x}_B, \bar{y}_B) = \left(0, 6 + \frac{4}{2}\right) = (0, 8)$$
   - **Volume V_B:**
   $$V_B = l_B^3 = 4^3 = 64$$
3. **Cone C:**
   - **Dimensions:** Diameter $l_C = 4 \implies$ radius $r_C = 2$. Height $h_C = \frac{3}{2} h_A = \frac{3}{2}(6) = 9$.

- **Position:** The circular base of diameter 4 is attached flat against one vertical 4×4 side face of Cube B. Taking this attachment face to be at $x = +2$ (spanning from $y = 6$ to $y = 10$), the cone extends horizontally outward along the positive x-direction from $x = 2$ to $x = 2 + 9 = 11$.
- **Center of mass** $(\bar{x}_C, \bar{y}_C)$**:** The vertical height of its axis aligns with the center of the face at $y = 8$. For a right circular cone, the centroid lies at a distance of $\frac{1}{4}$ of the height from the base along its symmetry axis:

$$\bar{x}_C = 2 + \frac{1}{4}h_C = 2 + \frac{9}{4} = \frac{17}{4} = 4.25, \qquad \bar{y}_C = 8$$

- **Volume** V_C**:**

$$V_C = \frac{1}{3}\pi r_C^2 h_C = \frac{1}{3}\pi(2^2)(9) = 12\pi$$

We calculate the total volume of the composite structure:

$$V_{\text{total}} = V_A + V_B + V_C = 24\pi + 64 + 12\pi = 36\pi + 64$$

We now compute the exact coordinates $(\bar{x}, \bar{y})$ of the overall center of mass:

- **Horizontal coordinate** $\bar{x}$**:**

$$\bar{x} = \frac{\bar{x}_A V_A + \bar{x}_B V_B + \bar{x}_C V_C}{V_{\text{total}}} = \frac{0 \cdot (24\pi) + 0 \cdot 64 + \left(\frac{17}{4}\right)(12\pi)}{36\pi + 64} = \frac{51\pi}{36\pi + 64}$$

- **Vertical coordinate** $\bar{y}$**:**

$$\bar{y} = \frac{\bar{y}_A V_A + \bar{y}_B V_B + \bar{y}_C V_C}{V_{\text{total}}} = \frac{3 \cdot (24\pi) + 8 \cdot 64 + 8 \cdot (12\pi)}{36\pi + 64} = \frac{168\pi + 512}{36\pi + 64}$$

Summing the coordinates gives:

$$\bar{x} + \bar{y} = \frac{51\pi + 168\pi + 512}{36\pi + 64} = \frac{219\pi + 512}{36\pi + 64}$$

We substitute this sum into the definition of the target parameter n:

$$n = 2(\bar{x} + \bar{y}) + 1 = 2\left(\frac{219\pi + 512}{36\pi + 64}\right) + 1 = \frac{438\pi + 1024 + 36\pi + 64}{36\pi + 64} = \frac{474\pi + 1088}{36\pi + 64}$$

Dividing numerator and denominator by 4:

$$n = \frac{118.5\pi + 272}{9\pi + 16}$$

Using the high-precision numerical value $\pi \approx 3.141592653589793$, we evaluate the fraction:

$$\text{Numerator} = 118.5(3.1415926536) + 272 = 372.27872945 + 272 = 644.27872945$$

$$\text{Denominator} = 9(3.1415926536) + 16 = 28.27433388 + 16 = 44.27433388$$

$$n = \frac{644.27872945}{44.27433388} \approx 14.55196889\ldots$$

Rounding to 5 decimal places, the value of n is:

$$n = 14.55197$$

4.2 Quantum Mechanics

Compton Backscattering Energy Calculation

A gamma ray initially has an energy of $E = 5.00$ MeV. It scatters from a stationary electron through an angle of 180°. Calculate the scattered photon energy E' in MeV. Use the electron rest mass energy $m_e c^2 = 0.511$ MeV.

Solution: The wavelength shift of a photon undergoing Compton scattering by a stationary free electron through a scattering angle θ is governed by the standard formula:

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

where λ and λ' are the initial and scattered photon wavelengths, h is Planck's constant, m_e is the electron rest mass, and c is the speed of light. Using the fundamental de Broglie-Planck relation linking photon wavelength and energy, $E = \frac{hc}{\lambda} \implies \lambda = \frac{hc}{E}$, we convert the wavelength shift equation directly into an energy representation:

$$\frac{hc}{E'} - \frac{hc}{E} = \frac{h}{m_e c}(1 - \cos\theta)$$

Dividing both sides of the equation by hc:

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2}(1 - \cos\theta)$$

For a backscattered gamma ray where the photon is reflected directly backward, the scattering angle is $\theta = 180°$. Evaluating the trigonometric factor:

$$1 - \cos(180°) = 1 - (-1) = 2$$

Substituting this factor into the energy conservation formula yields:

$$\frac{1}{E'} = \frac{1}{E} + \frac{2}{m_e c^2}$$

We now substitute the given physical parameters: initial photon energy $E = 5.00$ MeV and electron rest mass energy $m_e c^2 = 0.511$ MeV:

$$\frac{1}{E'} = \frac{1}{5.00} + \frac{2}{0.511}$$

Converting the terms into decimal fractions:

$$\frac{1}{5.00} = 0.200000\ \text{MeV}^{-1}$$

$$\frac{2}{0.511} \approx 3.913894\ \text{MeV}^{-1}$$

Summing the two contributions:

$$\frac{1}{E'} = 0.200000 + 3.913894 = 4.113894\ \text{MeV}^{-1}$$

Taking the reciprocal to solve for the final scattered photon energy E':

$$E' = \frac{1}{4.113894\ \text{MeV}^{-1}} \approx 0.2430787\ \text{MeV}$$

Expressed as an exact rational fraction in terms of the initial inputs:

$$E' = \frac{5.00 \times 0.511}{0.511 + 2(5.00)} = \frac{2.555}{10.511} = \frac{2555}{10511}\ \text{MeV} \approx 0.24308\ \text{MeV}$$

Rounding to three significant figures to match the precision of the provided parameters, the scattered photon energy is:

$$E' \approx 0.243\ \text{MeV}$$

4.3 Electromagnetism

Concentric Spherical Gauss Law Flux Ratio

A circular disc of radius R carries surface charge density $\sigma(r) = \sigma_0\left(1 - \frac{r}{R}\right)$, where σ_0 is a constant and r is the distance from the center of the disc. Electric flux through a large spherical surface that encloses the charged disc completely is Φ_0. Electric flux through another spherical surface of radius $0.25R$ and concentric with the disc is Φ. Find the ratio $\frac{\Phi_0}{\Phi}$.

Solution: According to Gauss's Law in integral form, the total net electric flux Φ_S passing through any closed bounding surface S in vacuum is directly proportional to the total electric charge enclosed within that volume:

$$\Phi_S = \oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$$

Therefore, the ratio of the electric fluxes through any two closed surfaces centered around a charge distribution is equal precisely to the ratio of the total electric charges enclosed within each respective surface:

$$\frac{\Phi_0}{\Phi} = \frac{Q_{\text{total}}}{Q(0.25R)}$$

where Q_{total} is the total charge contained on the entire disc of radius R, and $Q(r_0)$ is the charge contained within a concentric sphere of radius $r_0 = 0.25R = \frac{R}{4}$. Because the disc lies flat in a single plane passing through the origin, the intersection of a concentric sphere of radius r_0 with the plane of the disc is exactly a concentric circle of radius r_0. Thus, the charge enclosed by the sphere is simply the surface charge integrated over a circular disc of radius r_0.

We compute the general expression for the charge $Q(r_0)$ enclosed within an arbitrary radius $r_0 \leq R$ by integrating the radial surface charge density $\sigma(r)$ using differential polar area elements $dA = r\,dr\,d\theta$:

$$Q(r_0) = \int_0^{2\pi} d\theta \int_0^{r_0} \sigma(r) r\,dr = 2\pi\sigma_0 \int_0^{r_0} \left(1 - \frac{r}{R}\right) r\,dr$$

Expanding the polynomial integrand:

$$Q(r_0) = 2\pi\sigma_0 \int_0^{r_0} \left(r - \frac{r^2}{R} \right) dr = 2\pi\sigma_0 \left[\frac{r^2}{2} - \frac{r^3}{3R} \right]_0^{r_0} = 2\pi\sigma_0 \left(\frac{r_0^2}{2} - \frac{r_0^3}{3R} \right)$$

We first determine the total charge Q_{total} by evaluating this expression at the full disc radius $r_0 = R$:

$$Q_{\text{total}} = Q(R) = 2\pi\sigma_0 \left(\frac{R^2}{2} - \frac{R^3}{3R} \right) = 2\pi\sigma_0 R^2 \left(\frac{1}{2} - \frac{1}{3} \right) = 2\pi\sigma_0 R^2 \left(\frac{1}{6} \right) = \frac{\pi\sigma_0 R^2}{3}$$

Next, we calculate the charge enclosed within the concentric sphere of radius $r_0 = 0.25R = \frac{R}{4}$:

$$Q(0.25R) = 2\pi\sigma_0 \left[\frac{(R/4)^2}{2} - \frac{(R/4)^3}{3R} \right] = 2\pi\sigma_0 R^2 \left[\frac{1}{32} - \frac{1}{192} \right]$$

Finding a common denominator ($192 = 32 \times 6$) to simplify the fraction:

$$\frac{1}{32} - \frac{1}{192} = \frac{6-1}{192} = \frac{5}{192}$$

Substituting this back into the enclosed charge expression:

$$Q(0.25R) = 2\pi\sigma_0 R^2 \left(\frac{5}{192} \right) = \frac{5\pi\sigma_0 R^2}{96}$$

Finally, we take the ratio of the two enclosed charges to find the desired electric flux ratio:

$$\frac{\Phi_0}{\Phi} = \frac{Q_{\text{total}}}{Q(0.25R)} = \frac{\frac{\pi\sigma_0 R^2}{3}}{\frac{5\pi\sigma_0 R^2}{96}} = \frac{1}{3} \cdot \frac{96}{5} = \frac{32}{5} = 6.4$$

The exact numerical value of the flux ratio is:

$$6.4$$